Learning Maps

Cadence Training Services learning maps provide a comprehensive visual overview of the learning opportunities for Cadence customers. They provide recommended course flows as well as tool experience and knowledge levels to guide students through a complete learning plan. Learning Maps cover all Cadence® technologies and reference courses available worldwide. For course names, descriptions, and schedules, please select the Browse Catalog button at https://www.cadence.com/training.

Contents

- PCB Design and Analysis
- Custom IC, Analog, and RF Design
- Digital Design and Signoff
- System Design and Analysis
- IC Package Design and Analysis
- Mixed Signal Modeling and Verification
- Computational Fluid Dynamics
- Safety and Reliability Platform
- Tensilica® Processor IP
- Reality DC
- Onboarding

cādence®

PCB Design and Analysis Learning Map

Beginner → Advanced

Logic Design

- Allegro X Design Entry HDL Basics (0.5)
- Allegro® X Design Entry HDL Front-to-Back Flow (3)
- Allegro EDM Design Entry HDL Front-to-Back Flow (3)
- Allegro X System Capture Basics (1)
- Allegro X System Capture Front-to-Back Flow (2.5)
- OrCAD® X Capture (2)
- OrCAD CIS (1)
- OrCAD X Capture Constraint Manager PCB Flow (1)
- Analog Simulation with PSpice® (3)
- Analog Simulation with PSpice® using System Capture (3)
- Analog Simulation with PSpice® using Design Entry HDL (4)

PCB Design

- OrCAD X Presto Basic Techniques (3)
- Allegro X PCB Editor Basic Techniques (3.5)
  - Allegro X PCB Editor Intermediate Techniques (2.5)
  - Allegro X PCB Router Basics (2)
  - Allegro X PCB Editor Advanced Methodologies (1)
- Allegro X High-Speed Constraint Management (2)
- Allegro DesignTrue DFM (2)
- Allegro X Update Training NEW (2)
- Advanced Design Verification with the RAVEL Programming Language (2)

SI/PI Analysis

- Essential High-Speed PCB Design for Signal Integrity (3)
- PCB Design at RF – Multi-Gigabit Transmission, EMI Control, and PCB Materials (2)
- Sigrity Aurora (2)
- Sigrity PowerDC™ and OptimizePI™ (1)
- DC and Thermal Analysis with Celsius PowerDC (2)
- PDN and Voltage Ripple Analysis with Sigrity X OptimizePI and SystemPI (2)
- SystemSI for Parallel Bus and Serial Link Analysis (2)
- Model Generation and Analysis using PowerSI and Broadband SPICE (3)
- Clarity 3D Solver (2)
- Celsius Thermal Solver (1)

Library Development

- DE-HDL Library Development using DE-HDL (2)
- DE-HDL Library Development using Allegro X System Capture NEW (2)
- Allegro X EDM PCB Librarian NEW (2)
- Allegro Design Entry HDL SKILL® Programming Language (3)
- Allegro X PCB Editor SKILL Programming Language (3)

NEW New Course | # Number of days for instructor-led course | Accelerated Learning Path Course | Online Course Available | Digital Badge Available

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IC Package Design and Analysis Learning Map

Beginner → Advanced

IC Package Design

- Allegro® X Advanced Package Designer (4)
- Allegro Sigrity Package Assessment and Model Extraction (1)
- Advanced Design Verification with the RAVEL Programming Language NEW (2)
- Designing with Integrity 3D-IC (2)

SI/PI Analysis

- Sigrity Aurora (2)
- Sigrity PowerDC™ and OptimizePI™ (1)
- SystemSI for Parallel Bus and Serial Link Analysis (2)
- Model Generation and Analysis using PowerSI and Broadband SPICE (3)
- Clarity 3D Solver NEW (2)
- Celsius Thermal Solver (1)

Beginner → Advanced

NEW New Course | # Number of days for instructor-led course | Accelerated Learning Path Course | Online Course Available | Digital Badge Available

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Custom IC/Analog Design and Simulation Learning Map

Beginner → Advanced

Circuit Schematic and Design Creation

- Virtuoso Schematic Editor S1: Creating Design Schematics (1.5)
- Virtuoso Schematic Editor S2: Navigating and Capturing Design Intent (1.5)

Virtuoso® ADE Explorer & Assembler Series

- **S1** ADE Explorer & Single Test Corner Analysis (0.5)
- **S2** ADE Assembler & Multi Test Corner Analysis (1)
- **S3** Sweeping Variables and Simulating Corners (0.5)
- **S4** Monte Carlo, Real-Time Tuning & Run Plans (0.5)

- Analyzing Simulation Results Using Virtuoso Visualization and Analysis (1)

Spectre Simulations

Spectre® Simulator Fundamentals Series

- **S1** Spectre Basics (1)
- **S2** Large-Signal Analyses (1)
- **S3** Small-Signal Analyses (1)
- **S4** Spectre MDL (0.5)

- Design Checks and Asserts In Spectre Simulator (2.5)
- High-Performance Spectre Simulation (APS, Spectre X) (1)
- FastSpice Simulations Using Spectre FX Simulator (1)
- Spectre FMC in Virtuoso ADE Artist (1.5)

Using Spectre® Effectively Series

- **S1** Accelerating DC Analysis (0.5)
- **S2** Accelerating Transient Analysis (1)

- Virtuoso Spectre Transient Noise (2)

Design Verification

- Managing Analog Verification Using ADE Verifier (2)

Reliability Analysis

- Reliability Analysis in Virtuoso Studio (0.5)

Library Characterization

- Characterizing Standard Cells Using Liberate (1.5)

Accelerated Learning Path Course

NEW New Course | # Number of days for instructor-led course | L XL GXL Tiers of Cadence products used in course | Online Course Available | Digital Badge Available

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Custom IC/Analog Physical Design and Verification Learning Map

Beginner → Advanced

IC CAD

- SKILL® Language Programming Introduction (2)
  - SKILL Language Programming (5)
  - SKILL Development of Parameterized Cells (2)
  - Advanced SKILL Language Programming (3)

Photonics

- Virtuoso Layout for Photonics Design - T1 (NEW) (1)

Physical Design and Advanced Nodes

- Virtuoso® Layout Design Basics (NEW) (2)
  - Virtuoso Abstract Generator (EXL) (2)
  - Virtuoso Floorplanner (NEW) (EXL) (1)

Virtuoso® Advanced Nodes

- Virtuoso Layout for Advanced Nodes (2)
  - T1: Place and Route (1)
  - T2: Electromigration (0.5)
  - Auto Place and Route (APR) for Virtuoso Studio – Device Level (2)
  - Auto Place and Route (APR) for Virtuoso Studio – Standard Cell (NEW) (1)

- Cadence Analog IC Design Flow (NEW) (3)

Virtuoso® Layout Pro Series

- T1: Env. and Basic Commands (1)
- T2: Create and Edit Commands (1)
- T3: Basic Commands (1)
- T4: Advanced Commands (1)
- T5: Interactive Routing (1)
  - T6: Constraint-Driven Flow and Power Routing (EXL) (1)
  - T7: Module Generator and Floorplanner (EXL) (1)
- T8: Concurrent Layout Editing (EXL) (1)
- T9: Virtuoso Design Planner (EXL) (2)

- Virtuoso Studio Features (NEW) (2)

Physical Verification and Extraction

- Pegasus Verification System (NEW) (2)
  - Physical Verification System (2)
    - Physical Verification Language Rules-Writer (NEW) (2)
- Transistor-Level Power Signoff with Voltus-XFi (NEW) (2)

Quantus™ Extraction Solution Transistor-Level Series

- T1: Overview and Technology Setup (0.5)
  - T2: Parasitic Extraction (1)
  - T3: Extracted View Flows and Advanced Features (0.5)

Accelerated Learning Path Course

NEW New Course | # Number of days for instructor-led course | XL GXL EXL Tiers of Cadence products used in course | Online Course Available | Digital Badge Available

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RF/Microwave and System Design Learning Map

2 of 3 – see next page

Beginner → Advanced

Spectre RF Simulation and Analysis

S2 ADE Assembler & Multi Test Corner Analysis

Spectre Simulator Fundamentals **S2** Large-Signal Analyses

Large-Signal RF Analyses Using Harmonic Balance and Shooting Newton (NEW)

S2 ADE Assembler & Multi Test Corner Analysis

Spectre Simulator Fundamentals **S3** Small-Signal Analyses

Small-Signal RF Analyses Using Harmonic Balance and Shooting Newton (NEW)

5G mmWave Handset System Design – **S1** Simulation and Verification of the RFIC (Transceiver)

S-Parameter Simulations Using Spectre in Virtuoso Studio (NEW)

RF/Microwave System Design

Microwave & RF Design (AWR®)

Microwave Office for RF Designers

Planar EM Analysis in AWR Microwave Office

3D EM Analysis with Clarity in Microwave Office

RF Receiver Design Using Cadence VSS

Electromagnetic Analysis

Electromagnetic Simulations Using the EMX Simulator

Virtuoso RF Solution

Virtuoso Heterogeneous Integration: EM Analysis of ICs Using the EMX Solver

NEW New Course | # Number of days for instructor-led course | L XL GXL Tiers of Cadence products used in course | Online Course Available | Digital Badge Available

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Mixed-Signal Modeling, Simulation and Verification Learning Map

Beginner → Advanced

Circuit Modeling

AMS/Real Number Modeling

- Analog Modeling with Verilog-A (3)
- Behavioral Modeling with Verilog AMS (2.5)
- Real Modeling with Verilog-AMS (2)
- Real Modeling with SystemVerilog (3)

Mixed-Signal Implementation

- Virtuoso® Digital Implementation (3)

Mixed-Signal Simulations (GUI and Command-Line)

- Virtuoso Schematic Editor S1: Creating Design Schematics (1.5)
- Virtuoso Schematic Editor S2: Navigating and Capturing Design Intent (1.5)
- **S1** ADE Explorer & Single Test Corner Analysis (0.5)
- **S2** ADE Assembler & Multi Test Corner Analysis (1)

Mixed-Signal Design and Simulation

- Mixed-Signal Simulations using Spectre AMS Designer (GUI) (2)
- Command-Line Based Mixed-Signal Simulations w/ Xcelium Use Model (2)
- SimVision for Debugging Mixed-Signal Simulations (1)

Mixed-Signal Verification

- SystemVerilog for Design and Verification (5)
- SystemVerilog Real Number Modeling (SV-RNM) Based Advanced Verification (5)
- Mixed-Signal Verification with UVM (NEW) (1.5)

Beginner → Advanced

NEW New Course | # Number of days for instructor-led course | L XL GXL Tiers of Cadence products used in course | Online Course Available | Digital Badge Available

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Digital Design and Signoff Learning Map

1 of 2 – see next page

Beginner → Advanced

Synthesis and Test

Implementation

- Semiconductor 101 (0.25)
- Introduction to Electronic Design Automation NEW (0.5)
- Digital IC Design Fundamentals (2.25)
- Cadence® RTL-to-GDSII Flow (2)
- Fundamentals of IEEE 1801 Low-Power Specification Format (1)
- Stratus High-Level Synthesis (HLS) Basic Training (4)
- Genus™ Synthesis Solution with Stylus Common UI (M, 3)
- Genus Low-Power Synthesis Flow with IEEE 1801 (1)
- Low-Power Synthesis Flow with Genus Stylus Common UI (1)
- Genus Physical Synthesis Flow (0.5)
- Advanced Synthesis with Genus Stylus Common UI (2)
- Joules™ Power Calculator (2.5)
- Design For Test Fundamentals (0.5)
- Test Synthesis with Genus Stylus Common UI (2)
- ATPG Flow with Modus DFT Software Solution (1.5)
- Diagnostics with Modus DFT Software Solution NEW (0.5)
- RTL Design Optimization with Joules RTL Design Studio (1.5)
- Innovus™ Block Implementation with Stylus Common UI (M, 3)
- Innovus Hierarchical Implementation with Stylus Common UI (1)
- Innovus Low-Power Flow with Stylus Common UI (1)
- Innovus Clock Concurrent Optimization Technology with Stylus Common UI (1)
- Cadence® Certus™ Signoff Closure Solution with Stylus Common UI (0.5)
- Voltus InsightAI (0.5)
- Innovus+ Synthesis and Implementation Flow with Stylus Common UI (0.5)
- Virtuoso® Digital Implementation (3)
- Artificial Intelligence and Machine Learning Fundamentals (0.5)
- Cadence® Cerebrus™ Intelligent Chip Explorer (1)

NEW New Course | # Number of days for Instructor-led Course | Online Course Available | Digital Badge Available | Accelerated Learning Path Course | M Mandarin Transcript

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Digital Design and Signoff Learning Map

Beginner

Advanced

Silicon Signoff

Equivalence Checking

Semiconductor 101 (0.25)

Introduction to Electronic Design Automation NEW (0.5)

Digital IC Design Fundamentals (2.25)

Cadence® RTL-to-GDSII Flow (2)

Basic Static Timing Analysis (1)

Tempus™ Signoff Timing Analysis and Closure with Stylus Common UI (2)

Voltus™ Power Grid Analysis and Signoff with Stylus Common UI (2)

Cadence® Certus™ Signoff Closure Solution with Stylus Common UI (0.5)

Voltus InsightAI (0.5)

Conformal® Equivalence Checking (2)

Conformal Low-Power Verification with CPF (1)

Conformal Low-Power Verification Using IEEE1801 (1)

Conformal ECO (1)

Artificial Intelligence and Machine Learning Fundamentals (0.5)

Cadence® Cerebrus™ Intelligent Chip Explorer (1)

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System Design and Verification Learning Map

Beginner

Simulation, Coverage and Debug

- Digital IC Design Fundamentals — NEW — 2.25
- Xcelium™ Simulator — NEW — 2
- Protium™ Introduction — NEW — 1
- Palladium® Introduction — NEW — 1
- Cadence® RTL-To-GDSII Flow — 2
- Low-Power Simulation with CPF — 2
- Low-Power Simulation with IEEE1801 UPF — 2
- VIP Basic Building Blocks and Usage — NEW — 2
- UCIe VIP Introduction — NEW — 3
- Foundations of Metric-Driven Verification — 1
- Xcelium Integrated Coverage — 2
- Verisium™ Manager — NEW — 3
- vManager Tool Usage in Batch Mode — 0.5
- Verisium™ Debug — NEW — 1
- Perspec™ System Verifier - Basic — NEW — 3
- Xcelium Fault Simulator — NEW — 1.5
- Specman® Fundamentals for Block-Level Environment Developers — 5
- Specman Advanced Verification — 4

Advanced

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System Design and Verification Learning Map

Beginner

Design and Verification Languages

Verilog Language and Application (4)

VHDL Language and Application (5)

C++ Language Fundamentals for Design and Verification (3)

SystemC® Language Fundamentals (3)

Stratus High-Level Synthesis (HLS) Basic Training (4)

SystemC Transaction-Level Modeling TLM2.0 (2)

SystemVerilog for Design and Verification (5)

Real Number Modeling with Verilog-AMS (2)

Real Number Modeling with SystemVerilog (3)

SystemVerilog Real Number Modeling (SV-RNM) Based Advanced Verification (5) NEW

Perl for EDA Engineering (3)

Tcl Scripting for EDA (2) NEW

Introduction to TK (1)

UVM

Essential SystemVerilog for UVM (optional) (1)

SystemVerilog Accelerated Verification Using UVM (4)

SystemVerilog Advanced Register Verification Using UVM (2) NEW

SystemVerilog Assertions (2)

Jasper® Formal Fundamentals (2)

Jasper® Formal Expert (2) NEW

Advanced

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Safety and Reliability Platform Learning Map

Beginner

Midas Safety Platform

Midas™ Safety Platform Introduction NEW 0.5

Xcelium Fault Simulator NEW 1.5

Midas™ Safety Analysis Authoring NEW 0.5

Verisium Manager Safety Flow NEW 1

Functional Safety Implementation and Verification with Midas™ 0.5

Advanced

Beginner

Advanced

NEW New Course | # Number of days for instructor-led course | Accelerated Learning Path Course | Online Course Available | Digital Badge Available

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Tensilica Processor IP Learning Map

Tensilica Xtensa LX

- Tensilica® Xtensa® LX Processor Fundamentals
- Tensilica Xtensa LX Processor Interfaces
- Tensilica Xtensa LX Hardware Verification and EDA
- Tensilica Instruction Extension Language and Design
- Tensilica System Modeling using XTSC

ConnX DSP

- Tensilica ConnX DSP Family NEW
- Tensilica ConnX BBE32EP Baseband Engine

Fusion, FloatingPoint & MathX DSP

- Tensilica Fusion F1 DSP
- Tensilica Fusion G3 DSP
- Tensilica FloatingPoint DSP Family
- Tensilica MathX DSP Family NEW

HiFi Audio DSP

- Tensilica Audio Codec API
- Tensilica Xtensa Audio Framework
- Tensilica HiFi 3 Audio Engine ISA
- Tensilica HiFi 4 DSP
- Tensilica HiFi 5 DSP

Vision DSP

- Tensilica Vision DSP Family NEW

NEW New Course | # Number of days for instructor-led course | Online Course Available

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Tensilica Processor IP Learning Map

Tensilica Xtensa NX

Tensilica® Xtensa® NX Processor Fundamentals

- Tensilica Xtensa NX Processor Interfaces
- Tensilica Xtensa NX Hardware Verification and EDA
- Tensilica Instruction Extension Language and Design
- Tensilica System Modeling using XTSC

ConnX DSP

- Tensilica ConnX B10 DSP
- Tensilica ConnX DSP Family NEW

Vision DSP

- Tensilica Vision DSP Family NEW

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Computational Fluid Dynamics

Beginner → Advanced

CFD Academy

- CFD Online Course (2.75)

Fidelity

Turbomachinery

- Fidelity Turbo: Introduction (3)

Meshing

- Fidelity Automesh for Unstructured Meshing (2)
- Fidelity Pointwise Meshing Foundations (1.75)

Auto Aero

- Fidelity Flow for Aerospace and High-speed Applications (0.5)

Fine

Marine

- Fine Marine for Beginners (3)
- Fine Marine for Advanced Users (2)

Beginner → Advanced

NEW New Course | # Number of days for instructor-led course | Online Course Available | Digital Badge Available

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Reality DC Learning Map

Beginner

Advanced

Design

- Introduction to Data Hall Modeling (3)
- External Environment Modeling (2)
- Transient Cooling Failure NEW (0.5)
- Flow Network Modeling (0.5)

Insight

Beginner

Advanced

NEW New Course

Number of days for instructor-led course

Online Course Available

Digital Badge Available

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Onboarding Curricula

Beginner → Advanced

PCB Design

- PCB Layout Designer Onboarding (3.75)
- Schematic Capture for EEs Onboarding (6.5)
- SI/PI Engineer Onboarding (6.5)
- EE/PCB Layout Designers Onboarding (6.5)

Custom IC, Analog, and RF Design

- Analog Circuit Design and Simulation Onboarding (6.25)
- Virtuoso Layout Onboarding (5)

Mixed-Signal Simulation and Verification

- Analog-Mixed Signal Design Modeling Onboarding (10)

Digital Design and Signoff

System Design and Verification

- System Design and Verification, Digital Physical Design and Signoff Onboarding (4.4)

See also: https://www.cadence.com/en_US/home/training/bridging-the-learning-gap/onboarding-curricula.html

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